



Faculty of Engineering and Technology
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Wireless and Mobile Networks, ENCS5323

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*Project – Real-World Spectrum Sensing and Controlled Radio Frequency Analysis
Using Software Defined Radio*

In this project, students will design and implement a real-world spectrum sensing system using the ADALM-PLUTO Software Defined Radio (SDR) platform. Students are expected to explore the behaviour of real RF signals across licensed and unlicensed bands, characterize the spectrum through live measurements, and critically analyse the effect of in-band interference on wireless system performance. The project culminates in a technical report supported by empirical evidence, reproducible code, and a discussion session. Each group will be allocated two ADALM-PLUTO units for a one-week borrowing period.

The project consists of two components. In the first component, students shall develop a spectrum sensing system that captures live RF signal data across a set of frequency bands of their choosing, drawn from cellular downlink bands (GSM 900, GSM 1800, UMTS 2100) and the 2.4 GHz ISM band (2400–2500 MHz). The software shall allow the user to configure the frequency range to be scanned within each band. For example, a group may choose to scan the entire GSM 900 downlink band from 935 to 960 MHz to observe the full set of active base station carriers, or alternatively focus on a narrower sub-range such as 2412 to 2437 MHz within the ISM band to examine the spectral activity around specific Wi-Fi channels. For each selected band, the system shall produce and the report shall present and interpret:

- Power spectral density with clearly labelled frequency axes, power scales, and reference levels
- Evidence of channel structure, occupancy patterns, and temporal variability in the measured spectrum

For the second component, students shall generate a controlled interference signal within the 2.4 GHz ISM band and measure its quantifiable effect on an active wireless receiver operating in the same band. The objective is to empirically demonstrate the relationship between interference characteristics and receiver performance degradation, and to relate the observed behaviour to wireless and mobile networks.

Responsibility and Safety Notice: The interference signal generator must be operated exclusively within a controlled, indoor environment, directed only toward equipment owned and operated by the group, and for the minimum duration necessary to capture the required measurements (typically 30 to 60 seconds per configuration). Operation of the interference generator is the full and sole responsibility of the group. Any use beyond the scope of this academic exercise, including any transmission that affects equipment not owned by the group or persons not part of the experiment, is strictly prohibited. Students must ensure that the transmitted power level and antenna placement are appropriate to the scale of the experiment and do not extend beyond the immediate measurement environment.

The interference signal shall be characterised in terms of its spectral footprint, centre frequency, and bandwidth, and these parameters shall be varied systematically across experimental conditions, with objective performance metrics of the affected receiver recorded for each configuration. For example, a group may configure the interference signal to cover the full 20 MHz bandwidth of a specific Wi-Fi channel and measure the resulting degradation in throughput and latency — using tools such as continuous ping monitoring and network speed measurement — then repeat the experiment with a narrower interference bandwidth covering only half the channel, and compare the two outcomes across conditions.

Both components shall be operable independently and concurrently. When run in parallel, the second component generates a controlled interference signal within the 2.4 GHz ISM band while the first component simultaneously senses and displays the spectral impact of that signal in real time.

Each group shall submit a self-contained technical report documenting the full scope of the project. The report is limited to six pages excluding the cover page, using Arial or Times New Roman font at size 12 with standard margins. Given the page limit, students are expected to prioritise results and their critical discussion over implementation details; the emphasis should be on what was observed, what it means, and how it relates to wireless communication theory rather than on describing the code or system setup at length. The complete source code for both the sensing system and the interference generator shall be submitted as a single archive; the code should be organised, commented, and reproducible. Each group will participate in an individual discussion session; during the session, all group members are expected to be prepared to run several experiments, discuss and defend the results and their interpretation, explain the observed spectral behaviour, and relate findings to the theoretical foundations of the course.

Equipment allocation, submission deadlines, and discussion session schedules are all linked to the borrowing period. A dedicated sign-up sheet will be announced through which groups register their borrowing week; the report and code are due the following Friday at 11:59 pm, the discussion session is held on Saturday, and the equipment must be returned on Saturday so that the next group may collect it on Monday.

Spectrum use: All RF transmissions conducted as part of this project must comply with applicable regulations governing the use of the 2.4 GHz ISM band. The low-power, short-duration, indoor nature of the interference generation component is consistent with permitted experimental use; however, students bear full personal responsibility for ensuring that their transmissions do not interfere with any equipment or persons outside the scope of their own experiment.

Naming convention: Name reports as **ENCS5323Project_<X1>_<X2>_<X3>_<X4>.pdf** where X1, X2, X3, X4 are the student numbers of the group members. Submit the source code as a ZIP archive named identically. Reports and code archives are submitted as a reply to the project announcement message on Ritaj.

Groups consist of three or four students. Each group submits a single report and code archive. All group members are individually accountable for the full content of the submission and must be prepared to discuss any aspect of it during the discussion session.